

ORIGINAL ARTICLE

ToRCH “co-infections” are associated with increased risk of abortion in pregnant women

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ABSTRACT ToRCH infections (toxoplasmosis, rubella, cytomegalovirus and Herpes simplex virus) have long been known to be associated with bad obstetric outcomes. However, little information is available about the impact of ToRCH co-infections on the outcome of pregnancy. Hence, we tested the IgG and IgM antibodies to *Toxoplasma gondii*, Rubella, Cytomegalovirus and Herpes Simplex Virus among 81 pregnant women with abortion (case group) and 98 pregnant women with normal delivery (control group). In the single-infection model, only CMV-IgM seropositivity was significantly increased in case than control group (25.9% in case and 12.2 % in control, OR = 2.5, $P = 0.019$). In the co-infection model, 14 patterns were recognized, but two patterns were significantly increased in the case than the control group. Co-infection of *T. gondii* IgG + CMV IgM was 9.1-fold increased in the case than the control group (8.6% in the case and 1% in control, OR = 9.1; $P = 0.024$). Also, co-infection of *T. gondii* IgG + HSV IgG + CMV IgM was 7.7-fold increased in case than the control group (7.4% in case and 1 % in control, OR = 7.7; $P = 0.04$). Although the OR of other co-infections was higher in the case than the control group, the difference was not statistically significant. These findings indicate that ToRCH co-infections are associated with increased risk of abortion than single infection. Hence, the rates of co-infections should be considered in prenatal screening of ToRCH infections.

Key Words: abortion, co-infection, pregnancy, ToRCH

INTRODUCTION

ToRCH infections (toxoplasmosis, rubella, cytomegalovirus (CMV) and Herpes simplex virus (HSV)) are the most common causes of congenital infections worldwide (Shet 2011; Vergnano and Heath 2013). Primary infections of toxoplasmosis, CMV, rubella and HSV during pregnancy can bring wide ranges of clinical symptoms dependent on the stage of pregnancy. ToRCH infections during the early stages of pregnancy may result in congenital malformations, intrauterine growth restriction (IUGR) or fetal death.

Infections during the later stages of pregnancy may result latent (asymptomatic) in babies at birth, which may progress to signs of infection at a later stage (Vergnano and Heath 2013). ToRCH infections can also bring a wide range of clinical symptoms such as skin rashes, jaundice, hepatosplenomegaly, lymphadenopathy and intracranial calcifications (Vergnano and Heath 2013).

Congenital toxoplasmosis occurs predominantly at the first time during pregnancy (Paquet and Yudin 2013). Seroprevalence rates of toxoplasmosis in females of reproductive age or pregnant women reported a range between 10% and 75% in different parts of the world (reviewed in [Pappas et al. 2009]). The incidence rate of congenital toxoplasmosis is approximately 1.5 cases per 1000 live births, with a global annual incidence of 190 100 cases (see recent systematic review [Torgerson and Mastroiacovo 2013]). Rubella infection during pregnancy has a devastating consequence; defined as congenital rubella syndrome (CRS). Miscarriage and stillbirth is the most important sequel of CRS (Duszak 2009). The incidence of CRS is approximately 0.6–2.2 cases per 1000 live births (Best et al. 2005). CMV is an extremely common viral infection with an overall 50% seroprevalence among adults (Boeckh and Geballe 2011); however, the seroprevalence rates are even higher in some populations with low and middle-income countries (Shet 2011; Manicklal et al. 2013). Congenital and neonatal infection is one of the most important consequences of primary CMV infection during pregnancy. The risk of primary infection in seronegative mothers ranges between 0.7–4.1% (Gandhi and Khanna 2004) with an overall birth prevalence of 0.64% (Kenneson and Cannon 2007). Neonatal HSV infections are rare but are associated with serious sequelae if not recognized. Seroprevalence rates of HSV-1 and HSV-2 are significantly depending on sex, age, race and geographic distribution (Pinninti and Kimberlin 2014). Primary HSV infection occurs when a susceptible person (lacking pre-existing HSV-1 and HSV-2 antibodies) is exposed to HSV (Straface et al. 2012). neonatal HSV infection is an uncommon occurrence with an estimated incidence of 1.6 to 8.4 per 100 000 live births (reviewed in (Pinninti and Kimberlin 2014)).

It is very well documented that non-immune pregnant women (lacking pre-existing antibodies to ToRCH infections) have a higher risk of congenital infections. Hence, congenital infections predominantly occur in mothers that acquired these infections in the first time during pregnancy (Shet 2011; Vergnano and Heath 2013). However, several studies reported an association between ToRCH seropositivity with a history of bad obstetric outcomes (Turbadkar et al. 2003; Aljumaili et al. 2013; Ramana et al. 2013; Alsamarai et al. 2014; Malik et al. 2014; Ghasemi et al. 2015b). On the other hand, there is little information about the impact of ToRCH co-infections on the outcome of pregnancy. Hence, our study aims

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Received July 17, 2015; revised and accepted September 18, 2015.

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Table 1 Study population

Risk factors	Variables	Case (<i>n</i> = 81)	Control (<i>n</i> = 98)	<i>P</i> -value
Age	<20	3 (3.7%)	6 (6.1%)	0.55‡
	20–39	75 (92.6%)	86 (87.8%)	
	>40	3 (3.7%)	6 (6.1%)	
Education level	Less than diploma	30 (37.1)	47 (47.9)	0.14†
	Diploma and above	51 (62.9)	51 (52.1)	
Job	House holder	76 (93.8%)	95 (96.9%)	0.47‡
	Practitioner	5 (6.2)	3 (3.1)	
Contact to cat	Yes	20 (24.7%)	23 (23.5%)	0.8†
	No	61 (75.3%)	75 (76.5%)	
Raw/half-cooked meat consumption	Yes	56 (69.1%)	70 (71.4%)	0.7†
	No	25 (30.9%)	28 (28.6%)	
Raw vegetables§	Yes	46 (56.8)	63 (64.3)	0.3†
	No	35 (43.2)	35 (35.7)	

P < 0.05.

† χ^2 test.

‡Fisher's exact test.

§Washing with detergent or not.

to investigate association of ToRCH seropositivity (especially co-infection of ToRCH infections) and abortion in a set of case-control study in Tehran, Iran.

MATERIAL AND METHODS

Subjects: cases and controls

This case-control study was performed among 179 pregnant women in three maternity hospitals (Akbar Abadi, Mirza Kochak Khan and Hedayat) in Tehran, Iran, from 2012 to 2013.

Eighty-one pregnant women with spontaneous abortion were enrolled as the case group, and 98 pregnant women with normal delivery were enrolled as the control group. A questionnaire including demographic, epidemiological criteria and clinical symptoms was recorded in both the case and control groups by interview.

Serological evaluation

Five milliliters of venous blood samples were collected from each pregnant woman in both case and control groups. Serum were separated from the blood samples and stored at -20°C until use. Specific IgG and IgM antibodies to *Toxoplasma gondii*, Cytomegalovirus, Rubella and HSV were measured by the Enzyme-linked immunosorbent assay, with commercial ELISA kits (Pishtaz Teb, Tehran, Iran) according to the manufacturer's instructions and the optical density (OD) was read at 450 nm by the spectrophotometer ELISA reader (Awareness Technology INC Stat Fax-2100). Diagnostic criteria IgG and IgM was defined the upper limit of the standard 10 U/mL (Cut-off).

Statistical analysis

All data were analyzed by SPSS version 11.5 (SPSS, Chicago, IL, USA) using χ^2 , Fisher exact test (exact sig). The odd ratio (OR) and *P*-value were also calculated.

RESULTS

The mean age of pregnant women in the case and control groups was 28.2 and 28.6 years, respectively, in which the difference was

Table 2 History of abortion and among case and control groups

History of abortion	Variables	Case (<i>n</i> = 81)
Number of abortions	One	67 (82.7%)
	Two	12 (14.8%)
	Three	2 (2.5%)
Time of abortion	First trimester	40 (49.4%)
	Second trimester	36 (44.4%)
	Third trimester	5 (6.2%)

not statistically significant. As shown in Table 1, there are no statistically significant differences in demographic factors among the case and control groups. The details of risk factors for *T. gondii* infection (such as job, contact to cat, raw/half-cooked meat consumption and consumption of raw vegetables) in the case and control groups are summarized in (Supplementary Table S1). Table 2 demonstrates the history of abortion among the case group. As observed, most of the abortions occurred in the first trimester of pregnancy (49.4%, 40/81), while 44.4% (36/81) and 6.2% (5/81) of abortions occurred in the second and third trimester of pregnancy, respectively (Table 2).

Toxoplasma gondii IgG antibody was detected in 27.2% (22/81) of case and 28.6% (28/98) of control group (OR = 1.07, *P* = 0.8). *T. gondii* IgM antibody was detected in 1.2% and 2% of case and control groups respectively (OR = 1.6, *P* = 1) (Table 3). The seropositivity of CMV IgG antibody was found in 87.7% and 90.8% of case and control groups respectively (OR = 0.7, *P* = 0.49). CMV IgM seropositivity was significantly higher in case (25.9%) than control group (12.2%) (OR = 2.5, *P* = 0.019). Rubella IgG and IgM seropositivities were higher in control than case group. Rubella IgG antibody was detected in 75.3% versus 86.7% in case and control groups (OR = 0.46, *P* = 0.05). Rubella IgM antibody was found in 1.2% vs. 2.04% in the case and control groups, respectively (OR = 0.6, *P* = 1). HSV seropositivity was also higher in the control

Table 3 Seroprevalence of *Toxoplasma gondii*, cytomegalovirus (CMV), Rubella and Herpes simplex virus (HSV) among case and control groups. Odds ratio (OR with 95% confidence interval [CI])

Infections	Antibody	Case (n = 81)		Control (n = 98)		P	OR	CI
		Positive	Negative	Positive	Negative			
<i>T. gondii</i>	IgG	22 (27.2%)	59 (72.8%)	28 (28.6%)	70 (71.4%)	0.8†	1.07	0.5–2.07
	IgM	1 (1.2%)	80 (98.8%)	2 (2%)	96 (97.9%)	1‡	1.6	0.14–18.7
	IgG + IgM	1 (1.2%)	80 (98.8%)	2 (2%)	96 (97.9%)	–	–	–
CMV	IgG	71 (87.7%)	10 (12.3%)	89 (90.8%)	9 (9.2%)	0.49†	0.7	0.27–1.86
	IgM	21 (25.9%)	60 (74.1%)	12 (12.2%)	86 (87.8%)	0.019†	2.5	1.14–5.48
	IgG + IgM	20 (24.7%)	61 (75.3%)	11 (11.2%)	87 (88.8%)	–	–	–
Rubella	IgG	61 (75.3%)	20 (24.7%)	85 (86.7%)	13 (13.3%)	0.05†	0.46	0.21–1.009
	IgM	1 (1.2%)	80 (98.8%)	2 (2.04%)	96 (98%)	1‡	1.6	0.14–18.7
	IgG + IgM	1 (1.2%)	80 (98.8%)	1 (1.02%)	97 (98.98%)	–	–	–
HSV	IgG	64 (79%)	17 (21%)	84 (85.7%)	14 (14.3%)	0.23†	1.59	0.7–3.4
	IgM	0	81 (100%)	3 (3%)	95 (96.9%)	0.25‡	–	–
	IgG + IgM	0	81 (100%)	2 (2.04%)	96 (97.96%)	–	–	–

P < 0.05.

† χ^2 test.

‡Fisher's exact test.

Table 4 ToRCH co-infection among case (n = 81) and control (n = 98) groups. Odds ratio (OR with 95% confidence interval [CI])

Co-infections	Case	Control	P	OR	CI
Toxo IgG + Rubella IgG	18 (22.2%)	26 (26.5)	0.5†	0.79	0.3–1.5
Toxo IgG + CMV IgG	20 (24.7%)	27 (27.6%)	0.6†	0.86	0.4–1.68
Toxo IgG + CMV IgM	7 (8.6%)	1 (1.02%)	0.024‡	9.17	1.1–76.2
Toxo IgG + HSV IgG	18 (22.2%)	26 (26.5%)	0.5†	0.79	0.39–1.57
Rubella IgG + CMV IgG	51 (62.9%)	81 (82.7%)	0.003†	0.35	0.17–0.71
Rubella IgG + CMV IgM	11 (13.6%)	9 (9.2%)	0.35†	1.5	0.6–3.95
HSV IgG + CMV IgG	55 (67.9%)	76 (77.5%)	0.14†	0.61	0.31–1.19
HSV IgG + CMV IgM	15 (18.5%)	9 (9.2%)	0.06†	2.2	0.92–5.44
Toxo IgG + CMV IgG + Rubella IgG	16 (19.7%)	26 (26.5%)	0.2†	0.68	0.33–1.38
Toxo IgG + CMV IgM + Rubella IgG	1 (1.2%)	5 (5.1%)	0.2‡	0.23	0.02–2.03
Toxo IgG + HSV IgG + CMV IgG	16 (19.8%)	25 (25.5%)	0.36†	0.71	0.35–1.46
Toxo IgG + HSV IgG + CMV IgM	6 (7.4%)	1 (1.02%)	0.047‡	7.7	0.9–65.8§
HSV IgG + Toxo IgG + CMV IgG + Rubella IgG	14 (17.3%)	24 (24.5%)	0.2†	0.64	0.3–1.3
HSV IgG + Toxo IgG + CMV IgM + Rubella IgG	8 (9.9%)	8 (8.2%)	0.6†	1.2	0.4–3.4

P < 0.05.

† χ^2 test.

‡Fisher's exact test.

§In this case, the SPSS does not make the correct 95% confidence interval result coordinated with P-value.

than the case group. HSV IgG seropositivity was found in 79% and 85.7% of the case and control groups (OR = 1.59, $P = 0.23$). HSV IgM was not detected in the case group, but detected in 3% of the control group ($P = 0.25$), (Table 3).

As reported in Table 3, CMV IgM seropositivity was 2.5 times higher in the case than the control group (25.9% in case and 12.2% in control, OR = 2.5, $P = 0.019$). Interestingly, the results indicated that the odds ratio of CMV IgM seropositivity co-infected with other infections was 1.2–9.1-fold higher in the case than the control

group (Table 4). While, *T. gondii* IgG + CMV IgM seropositivities were 9.1-fold higher in the case than the control group (8.6% in case and 1% in control, OR = 9.1; $P = 0.024$), HSV IgG + CMV IgM seropositivities were 2.2-fold higher in the case than the control group (18.5% vs 9.2%, OR = 2.2; $P = 0.06$), Rubella IgG + CMV IgM seropositivities were 1.5-fold higher in the case than the control group (13.6% vs 9.2% OR = 1.5; $P = 0.35$). Moreover, the rate of *T. gondii* IgG + HSV IgG + CMV IgM seropositivities were 7.7-fold higher in the case than the control group (7.4% vs 1%

OR = 7.7; $P = 0.04$). The rate of HSV IgG + *T. gondii* IgG + CMV IgM + Rubella IgG seropositivities were also 1.2-fold higher in the case than the control group (9.9% vs 8.2%, OR = 1.2; $P = 0.6$) (Table 4).

Clinical symptoms including fever, rash and lymphadenopathy were observed in both groups, although these rates were higher in the case than the control group. As observed in Figure 1, the rate of fever was significantly higher in the case group (29.6%, 24/81) than the control group (15.3%, 15/98) (OR = 2.3, $P = 0.021$). Rash was observed in 23.5% (19/81) of the case group and 16.3% (16/98) of the control group (OR = 1.5, $P = 0.23$). Moreover, four pregnant women in the case group (4.9%) and two (2%) in the control group had lymphadenopathy (OR = 2.4, $P = 0.41$). The details of clinical symptoms among *T. gondii*, CMV, Rubella and HSV infected individuals in the case and control groups are demonstrated in Table 5. Except in HSV IgM seropositive women, the rates of fever, rash and lymphadenopathy in IgG and IgM seropositivity were more increased in the case group than the control group (Table 5). The risk factors for *T. gondii* seropositivity among the case and control groups are demonstrated in supplementary Table S1.

DISCUSSION

Association of ToRCH seropositivity with a history of bad obstetric outcomes has been reported in several studies (Turbadkar et al. 2003; Aljumaili et al. 2013; Ramana et al. 2013; Alsamarai et al. 2014; Malik et al. 2014; Ghasemi et al. 2015b). However, there is

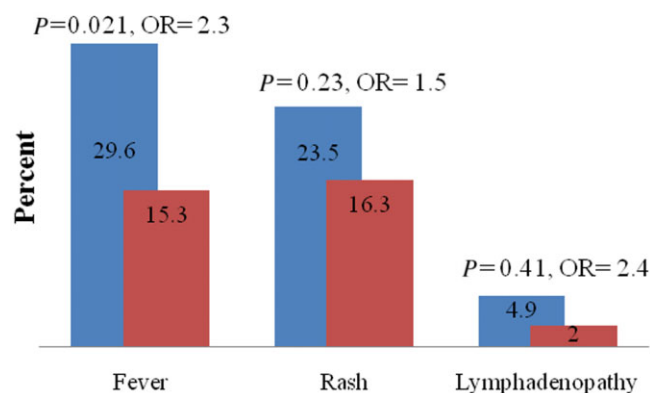


Fig. 1 Clinical symptoms among case and control groups. ■, Case ($n = 81$); ■, Control ($n = 98$).

little information describing the impact of ToRCH co-infections on the outcome of pregnancy. In the current study, the prevalence and co-occurrence of ToRCH infections (Table 3 and 4) along with demographic factors (Table 1) and some clinical symptoms (Table 2, 5 and Fig. 1) were measured in 81 pregnant women with spontaneous abortion as the case group and 98 pregnant women with normal delivery as the control group. We did not find any statistically significant differences in demographic factors among the case and control groups (Table 1). Most of the abortions occurred in the first (49.4%, 40/81) and second trimester of pregnancy (44.4%, 36/81) (Table 2). When we assessed each ToRCH infection, only CMV IgM seropositivity was significantly higher in the case than the control group (25.9% in case and 12.2% in control, OR = 2.5, $P = 0.019$). Also, Rubella IgG seropositivities were higher in the control than the case group (86.7% vs 75.3%, OR = 0.46, $P = 0.05$) (Table 3) probably due to the nationwide measles-rubella mass vaccination program (Honarvar et al. 2013). Interestingly, when we assessed ToRCH co-infections, the odds ratio of CMV IgM antibody positivity that co-infected with other infections was 1.2–9.1-fold increased in the case than the control group (Table 4). The predominant co-infection was *T. gondii* IgG + CMV IgM with OR = 9.1, followed by *T. gondii* IgG + HSV IgG + CMV IgM seropositivities with OR = 7.7, HSV IgG + CMV IgM seropositivities with OR = 2.2, Rubella IgG + CMV IgM seropositivities with OR = 1.5 and HSV IgG + *T. gondii* IgG + CMV IgM + Rubella IgG seropositivities with OR = 1.2 (Table 4). In this regard, Alsamarai et al. (2014) reported the rates of ToRCH co-infections in women with bad obstetric outcome and control group. They found a significant increased seroprevalence of *T. gondii* + CMV ($P = 0.038$), *T. gondii* + rubella + CMV ($P = 0.001$) and *T. gondii* + HSV-2 + CMV ($P = 0.015$) in women with bad obstetric outcome than the control group.

More than one-third of the world human population is infected with *T. gondii*, the most common form of infection of which is latent (asymptomatic) (Dalimi and Abdoli 2012). *T. gondii* infection can increase risk of adverse pregnancy outcomes (Li et al. 2014). Several studies reported an association between *T. gondii* infection and bad obstetric outcome (Turbadkar et al. 2003; Surpam et al. 2006; Kishore et al. 2011; Malik et al. 2014; Ghasemi et al. 2015b). The overall seroprevalence rates of toxoplasmosis are 39.9% (95% CI: 26.1–53.7) among childbearing age women in Iran (Borna et al. 2013). In the current study, *T. gondii* IgG and IgM antibodies were detected in 27.2% and 1.2% of the case group, while IgG and IgM seropositivities were observed in 28.6% and 2% of the control group, respectively. These rates were not significantly different (Table 3). Also the rates of fever, rash and lymphadenopathy were higher in *T. gondii* infected individuals in the case than the

Table 5 Clinical symptoms among *Toxoplasma gondii*, cytomegalovirus (CMV), Rubella and HSV seropositive individuals among case ($n = 81$) and control ($n = 98$) groups

Symptoms		Toxo		Rubella		CMV		HSV	
		IgG	IgM	IgG	IgM	IgG	IgM	IgG	IgM
Fever	Case	8 (9.9%)	1 (1.2%)	25 (30.9%)	1 (1.2%)	27 (33.3%)	9 (11.1%)	27 (33.3%)	0
	Control	1 (1%)	0	10 (10.2%)	0	12 (12.2%)	2 (2%)	10 (10.2%)	1 (1.02%)
Rash	Case	10 (12.3%)	1 (1.2%)	16 (19.8%)	0	18 (22.2%)	4 (4.9%)	16 (19.8)	0
	Control	3 (3.1%)	1 (1%)	9 (9.2%)	1 (1%)	8 (8.2%)	0	8 (8.2%)	1 (1.02%)
Lymphadenopathy	Case	3 (3.7%)	0	4 (4.9%)	1 (1.2%)	4 (4.9%)	1 (1.2%)	2 (2.4%)	0
	Control	0	0	0	0	1 (1%)	0	0	0

control group (Table 5). Based on the results of Ghasemi et al. (2015a,b), only the *T. gondii* IgM seropositivity was significantly higher in women with abortion (3.6%) than normal delivery women (0.9%) ($P = 0.37\%$). Also, Ghasemi et al. (2015a) found that the molecular prevalences of *T. gondii* were seven (6.4%) and two (1.8%) among aborted and normal delivery women, respectively.

We observed that the rate of *T. gondii* infection was associated with contact to cat, raw/half-cooked meat consumption and raw vegetables use in both the case and control groups. Higher infection rate was also observed in housekeeper women than practitioners (Supplementary Table S1). These observations have long been known to be risk factors of *T. gondii* infection (Pappas et al. 2009; Dalimi and Arshad 2011; Paquet and Yudin 2013).

Non-immune pregnant women are at risk of congenital rubella infection. A rubella vaccination program was initiated in December 2003 in Iran (Honarvar et al. 2013). Rubella vaccination has significantly reduced congenital rubella syndrome (CRS) in many developed countries and in some developing countries (Muller et al. 2007; Pandolfi et al. 2009). The prevalence of CRS in Iran was estimated to be 0.2 cases per 1000 live births before rubella vaccination (Sadighi et al. 2005), but there is not precise information about CRS prevalence at the present time in Iran. In the present study, higher prevalence (but not significant) of Rubella seropositivities (both IgG and IgM) was observed in the control than the case group, most likely due to the probably nationwide measles-rubella mass vaccination program (Honarvar et al. 2013). Rubella IgG and IgM antibodies were detected in 75.3% and 1.2% of the case group. Rubella IgG and IgM seropositivities were detected in 86.7% and 2.04% of the control group, respectively (Table 3).

Human CMV is an extremely common infection with an overall 50% seroprevalence among adults (Boeckh and Geballe 2011); however, the seroprevalence rates are even higher in some populations in low and middle-income countries (Shet 2011; Manicklal et al. 2013). Congenital infection is one of the most important consequences of primary CMV infection during pregnancy. The risk of congenital infection is greater in the first trimester than the second or third trimester of pregnancy (Gandhi and Khanna 2004). As estimated, the overall birth prevalence of congenital CMV infection is 0.64% (Kenneson and Cannon 2007), and the risk of primary CMV infection in seronegative mothers is in the range between 0.7 and 4.1% (Gandhi and Khanna 2004). CMV IgG and IgM prevalence has been reported in a range of 23–100% and 2.5–8.0% among childbearing age and pregnant women from different regions of Iran (Janbakhsh et al. 2001; Siadati et al. 2002; Arabzadeh et al. 2007; Armaghan et al. 2013). In the current study, the seropositivity of CMV IgG antibody was not significantly different in the case and control groups ($OR = 0.7$, $P = 0.49$). However, a significant increased in prevalence of CMV IgM antibody was observed in the case group (25.9%) than the control group (12.2%) ($OR = 2.5$, $P = 0.02$) (Table 3).

Neonatal HSV infections are uncommon but are associated with serious consequences if not recognized (Pinninti and Kimberlin 2014). Most neonatal infections result from exposure to HSV in a susceptible person through the genital tract during delivery (Corey and Wald 2009). As estimated, the incidence of neonatal HSV infection ranges between 1.6 and 8.4 per 100 000 live births [reviewed in (Pinninti and Kimberlin 2014)]. HSV seroprevalence has been reported in a range of 58–79% for HSV-1 and 3.5–63% for HSV-2 in Iran (Tayyebi and Sharifi 2010; Rezaei-Chaparpordi et al. 2012; Barazesh et al. 2014). In our study, HSV antibodies were not significantly higher in control than case group (Table 3).

In the current study, we observed that clinical symptoms including fever, rash and lymphadenopathy were higher in the case than their control group (Fig. 1), but only the rate of fever was significantly higher in the case and control groups ($P = 0.02$). We also observed that the percentages of clinical symptoms were higher in all infected women in the case than the control group, even in Rubella and HSV infection, in which the seroprevalence was higher in the control than the case group (Table 5).

It is well documented that pregnancy is associated with increased susceptibility and severity toward infectious diseases due to several hormonal and immunological alterations that occurs during the course of pregnancy (Krishnan et al. 2013; Sappenfield et al. 2013; Abdoli and Pirestani 2014; Kourtis et al. 2014). On the other hand, several studies indicated that co-infections have greater adverse outcomes than single infections (reviewed by (Griffiths et al. 2011; Pasman 2012; Abdoli and Pirestani 2014). Hence, it is reasonable that co-infections have greater adverse impacts than single infections during pregnancy and our study is consistent with this hypothesis.

Taken together, the results of this study provide a new insight about the role of ToRCH co-infections in the etiology of spontaneous abortion. These findings can be used for the design of prevention programs for ToRCH infections in pregnant women.

ACKNOWLEDGMENTS

The authors acknowledge the cooperation of the maternity clinics in Tehran, and department of Microbiology. The authors also grateful to the vice chancellor in research affairs of Kashan University of Medical Sciences for the financial support, grant no 9296.

COMPETING INTERESTS

None declared.

Ethical approval: The study was approved by the Ethics Committee of Kashan University of Medical Science.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1 Risk factors for *T. gondii* seropositivity among case and control groups.